

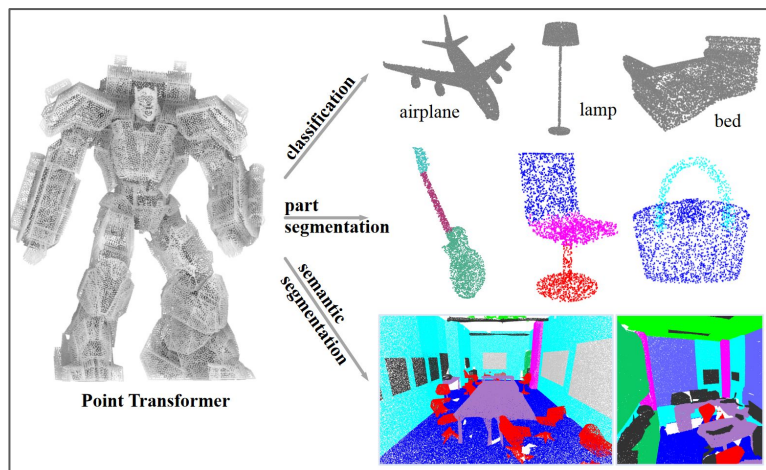
Deep Learning

Course Project AI 5100 (2024)

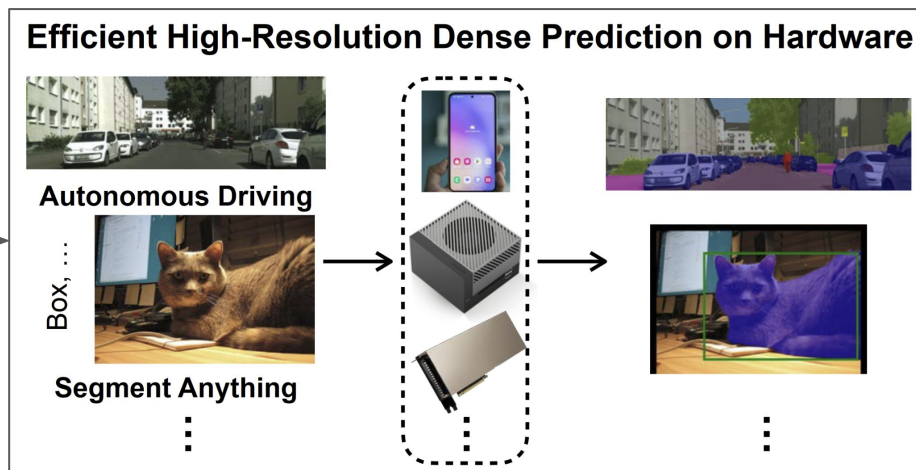
Ghanendra Singh, AI, IITH

Problem Statement

To build an efficient vision point transformer running in real time on edge devices.



Ref: Point Transformer



Ref: EfficientVit

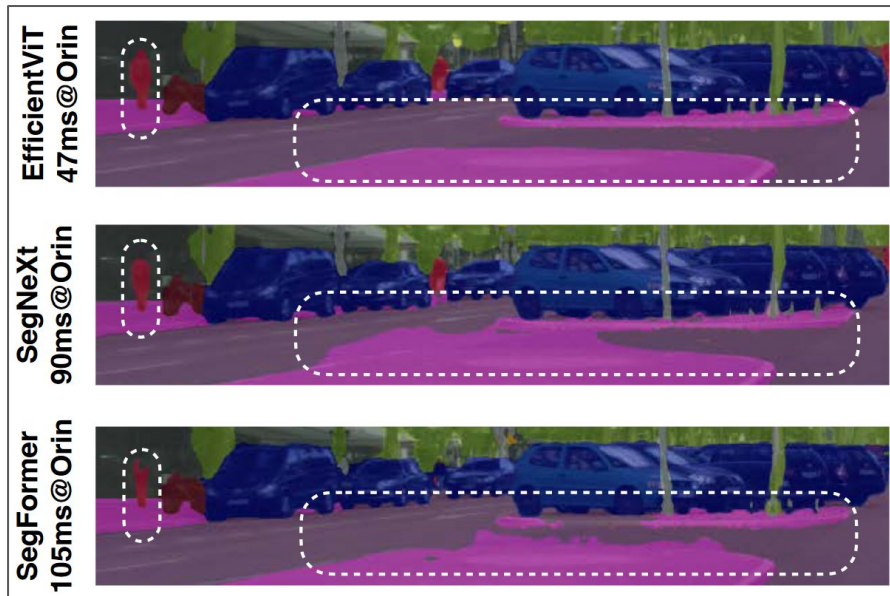
Objective: **eViT + PT** $\rightarrow$ **eVPT**

Efficient-ViT

A new family of vision transformer models for efficient high-resolution dense prediction.

Objective

A lightweight multi-scale attention module that simultaneously achieves a global receptive field, and multi-scale learning with lightweight and hardware-efficient operations, thus providing significant speedup on diverse hardware devices without performance loss than SOTA high resolution dense prediction models.



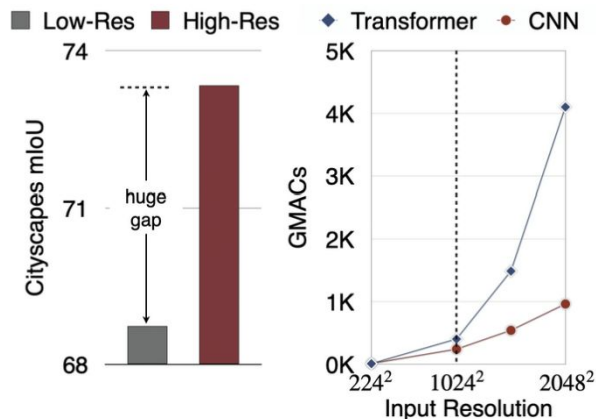
EfficientViT Street Scene Segmentation Demo

Running on NVIDIA Jetson AGX Orin (edge AI device) on Cityscapes



Challenge: Apply Transformer to High-Resolution Images

Details and small objects are missing in low-resolution images.

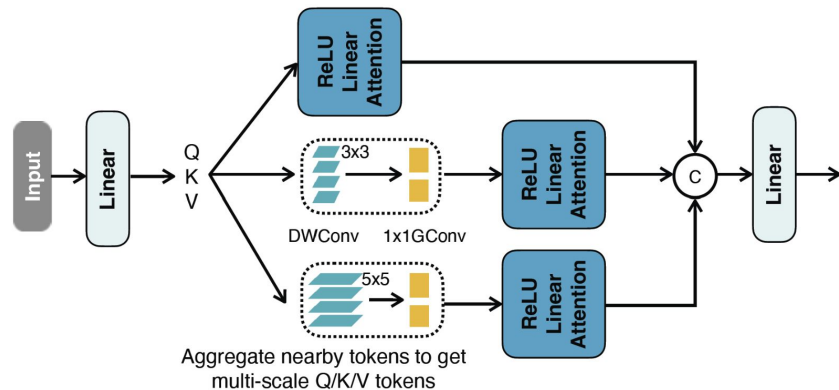
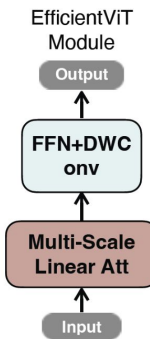


- High resolution is essential for achieving good performances in dense prediction tasks.
- Transformer's computational cost grows quadratically as the input resolution increases.

Efficient-ViT

High resolution dense prediction models require two critical features.

- Multiscale learning
- Global receptive field (GRF)



Hardware efficiency is essential for real-world applications.

- By replacing the inefficient softmax attention with lightweight ReLU linear attention
- ReLU linear attention reduce the computational complexity from quadratic to linear while preserving functionality.
- Multi-scale linear attention module to address the capacity limitation of ReLU linear attention.

Efficient ViT Building Block and Multi Scale Linear Attention

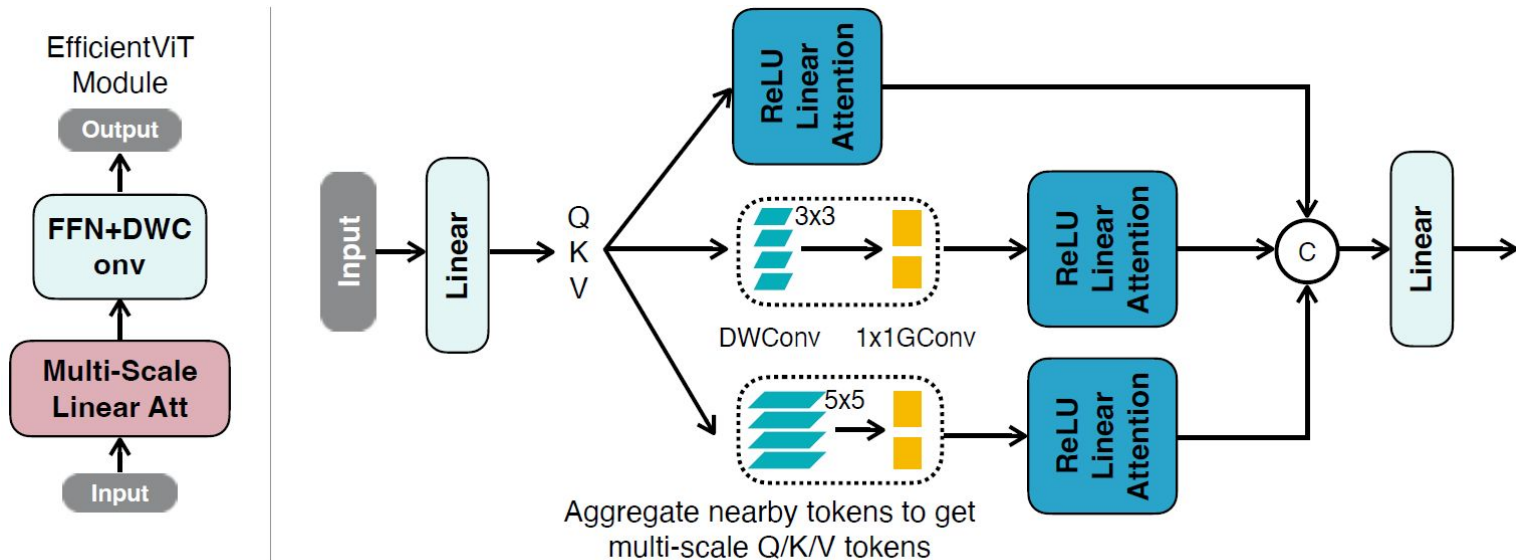


Figure 2: **EfficientViT's Building Block (left) and Multi-Scale Linear Attention (right).** *Left:* EfficientViT's building block consists of a multi-scale linear attention module and an FFN with depthwise convolution (FFN+DWConv). Multi-scale linear attention is responsible for capturing context information, while FFN+DWConv captures local information. *Right:* After getting Q/K/V tokens via the linear projection layer, we generate multi-scale tokens by aggregating nearby tokens via lightweight small-kernel convolutions. ReLU linear attention is applied to multi-scale tokens, and the outputs are concatenated and fed to the final linear projection layer for feature fusing.

EfficientViT

Attention Comparison

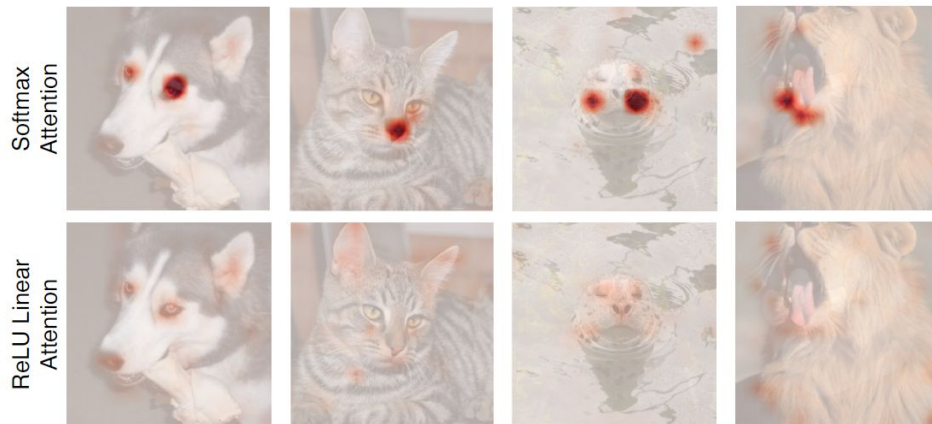
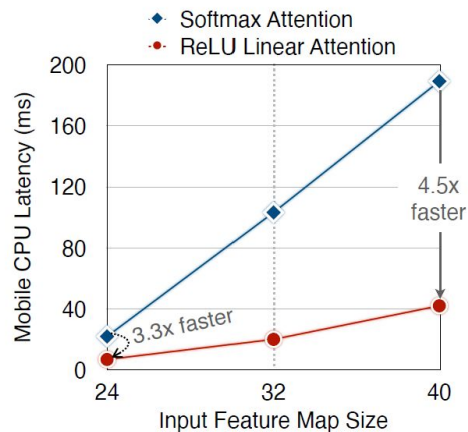
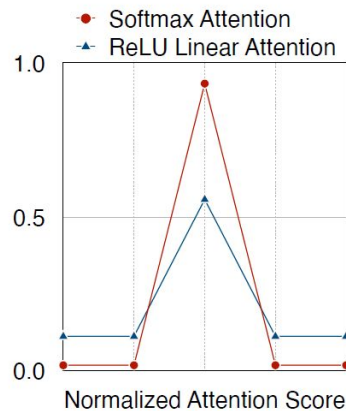


Figure 3: **Softmax Attention vs. ReLU Linear Attention.** Unlike softmax attention, ReLU linear attention cannot produce sharp attention distributions due to a lack of the non-linear similarity function. Thus, its local information extraction ability is weaker than the softmax attention.

Latency Comparison



EfficientViT Macro Architecture

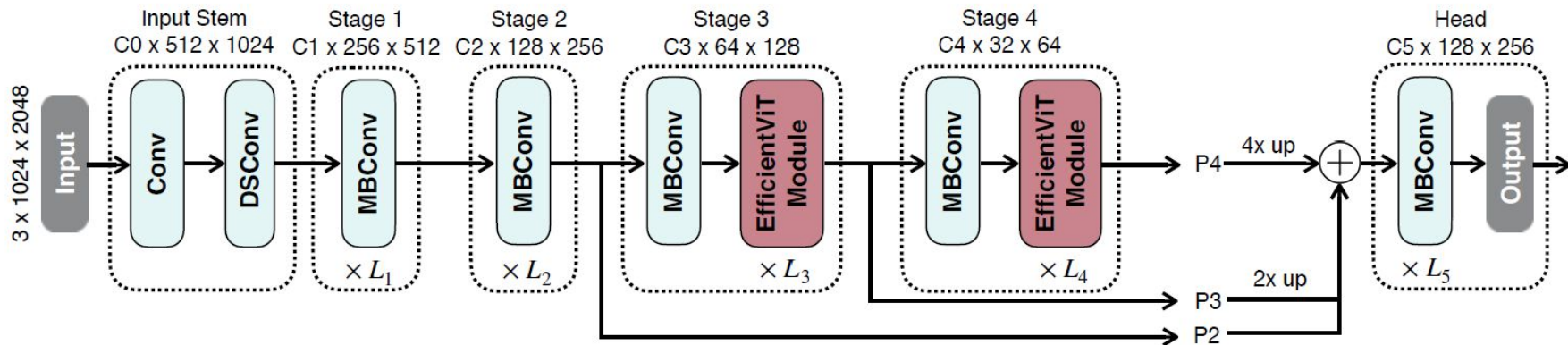
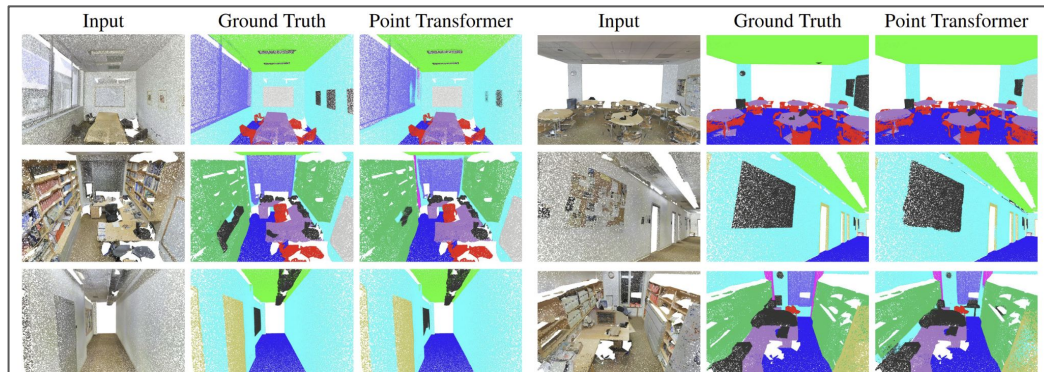


Figure 5: **Macro Architecture of EfficientViT.** We adopt the standard backbone-head/encoder-decoder design. We insert our EfficientViT modules in Stages 3 and 4 in the backbone. Following the common practice, we feed the features from the last three stages (P2, P3, and P4) to the head. We use addition to fuse these features for simplicity and efficiency. We adopt a simple head design that consists of several MBConv blocks and output layers.

Point Transformer

Self-attention networks to various 3D point cloud data processing for tasks such as semantic scene segmentation, object part segmentation, and object classification on point cloud data.

V1



V2

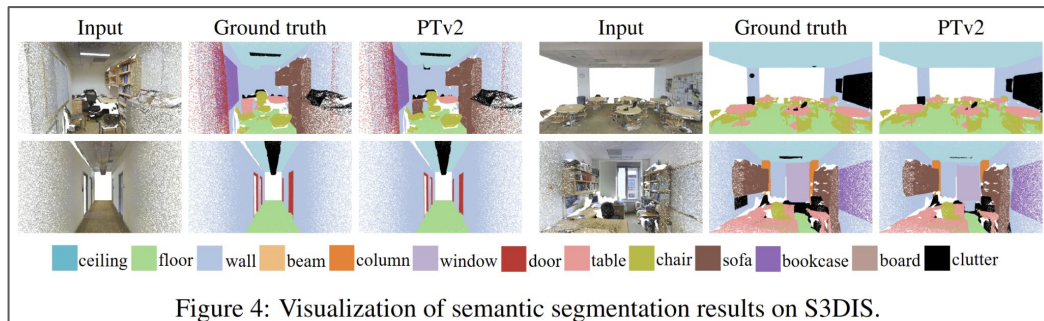


Figure 4: Visualization of semantic segmentation results on S3DIS.

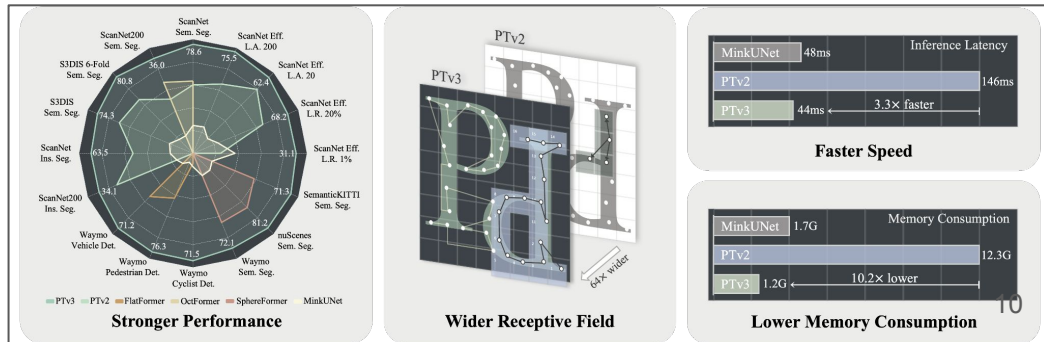
References

[Point Transformer](#) (ICCV 2021)

[Point Transformer V2](#) (NeurIPS 2022)

[Point Transformer V3](#) (CVPR 2024)

V3



Cityscapes Dataset: Depth Metadata

Sample PCD.

PCD processing is computationally very expensive for autonomous vehicles.

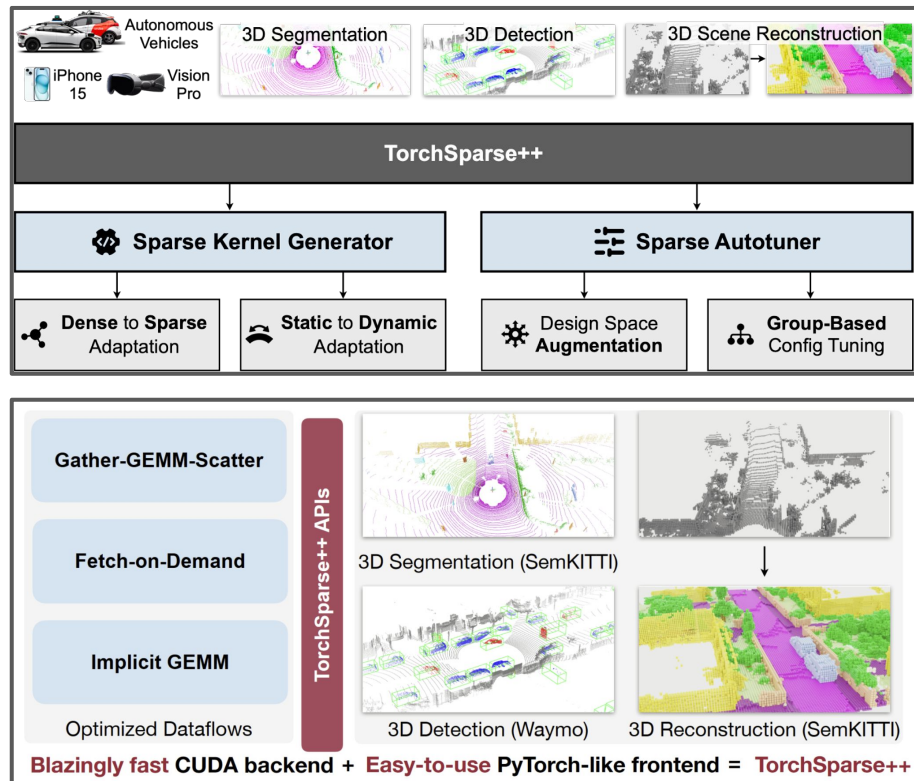
Any possible other method of depth?



Torch Sparse: Efficient Point Cloud Engine

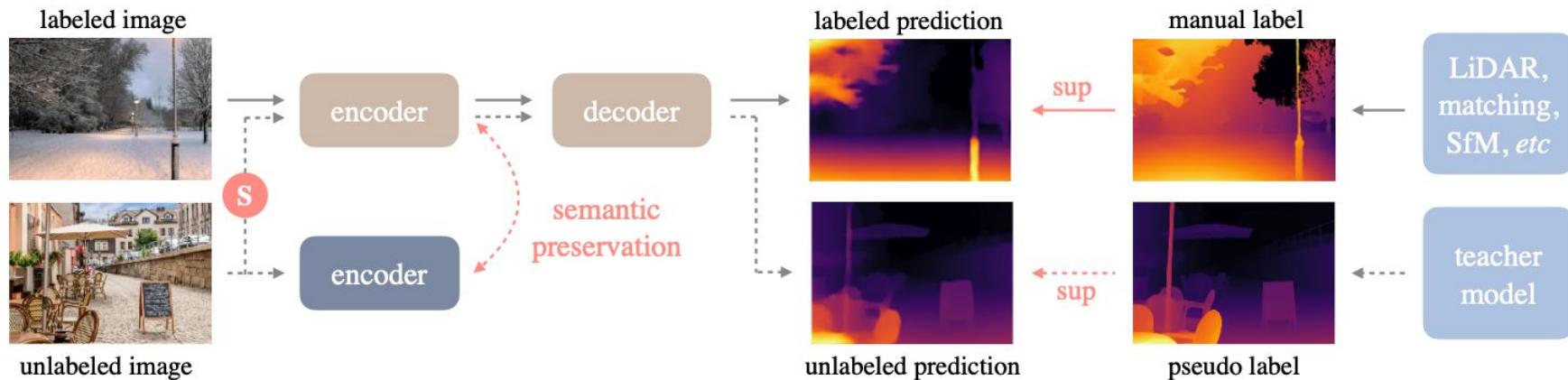
TorchSparse++

It is a high-performance library that provides highly-optimized dataflows for convolution on point clouds. It provides state-of-the-art inference and training performance for all driving-related applications (e.g. 3D semantic segmentation, object detection and scene reconstruction).



Depth Anything Model (TikTok)

Advancement in monocular depth perception.



Depth Anything Demo

Underwater marine life



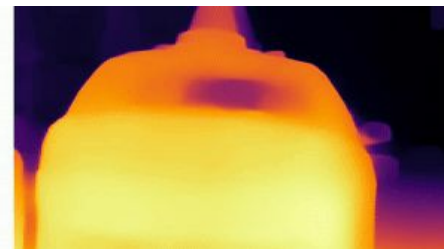
Human Action Observation



Gesture Recognition



Urban Environments



Results on KITTI and DDAD

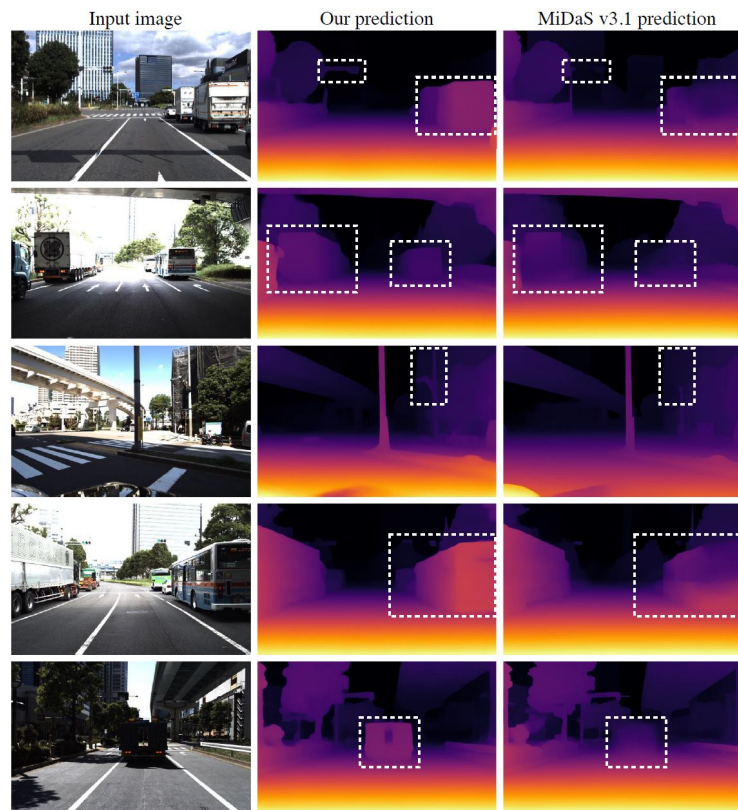
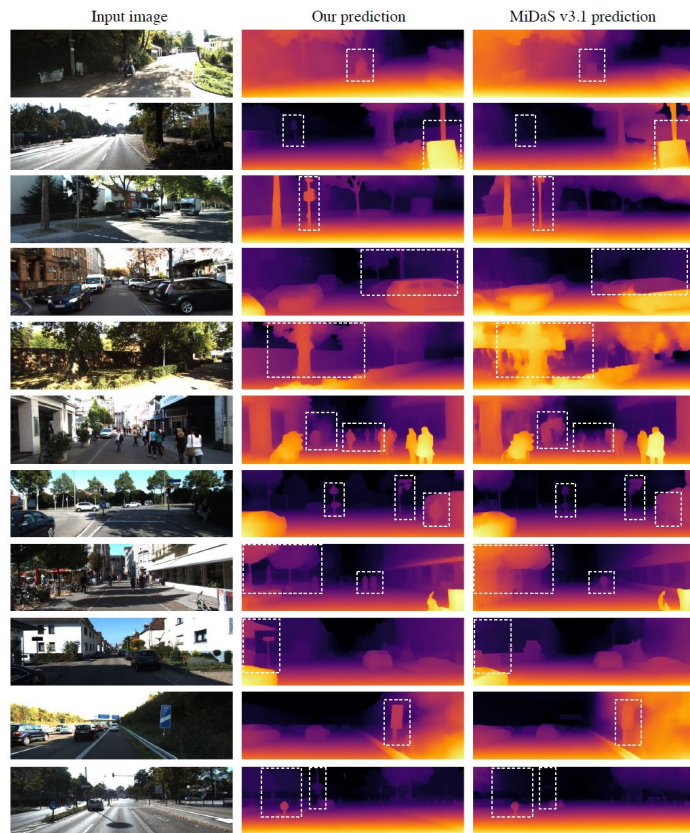


Figure 8. Qualitative results on DDAD.



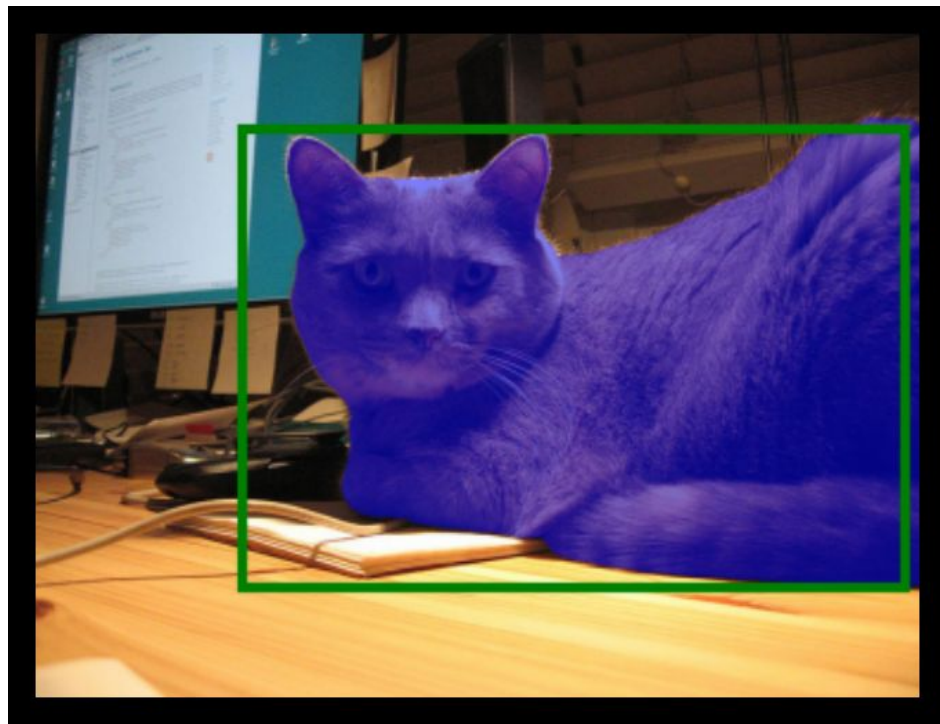
Methodology

Training a vision point transformer using components from efficient ViT and PT.

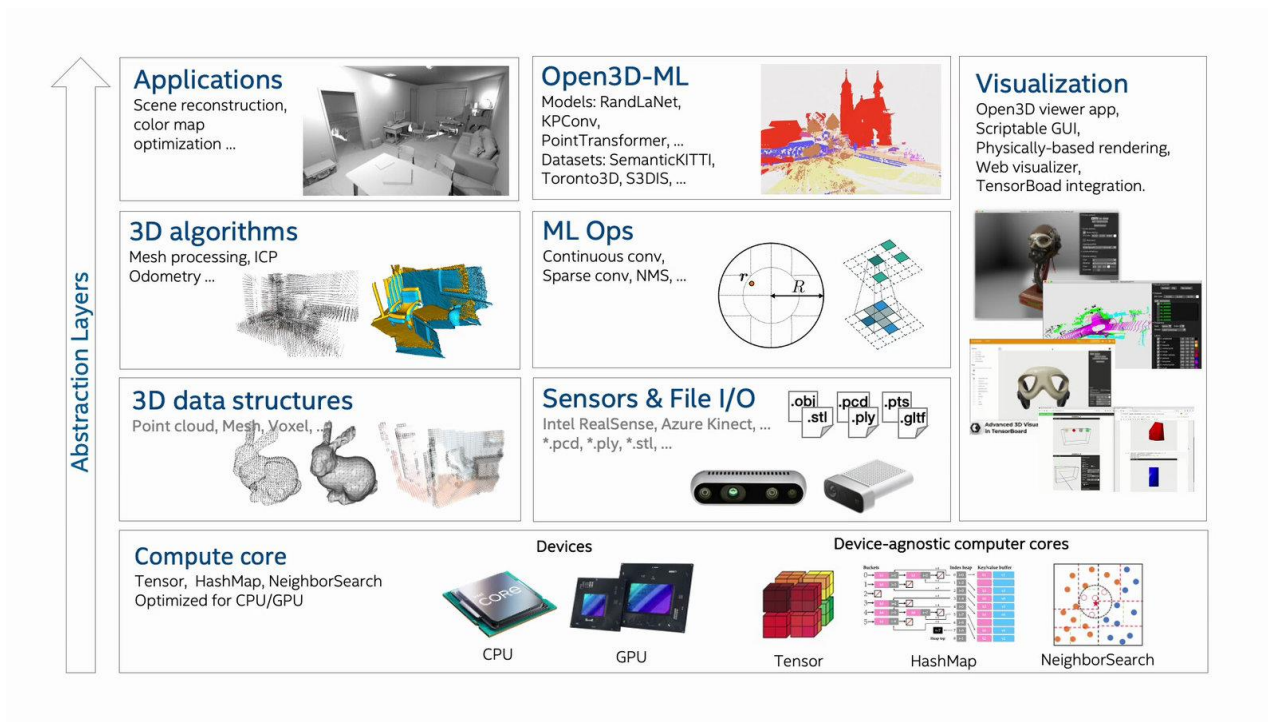
- Understanding point cloud data (PCD) and library for classical operations.
- Architecture and components of efficient Vision transformer (ViT).
- Architecture of Point Transformer (PT) across all versions (V1,V2, and V3)
- Train a Point transformer on available open source dataset.
- Collect dual channel stacked image and point cloud data.
- Building an improved version of combined vision point transformer.

Results

Test & run efficient Vit SAM benchmark
on Jetson Orin Nano.



Open3D Library



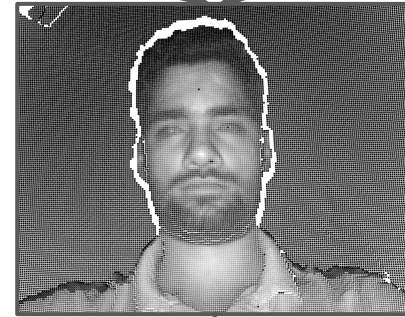
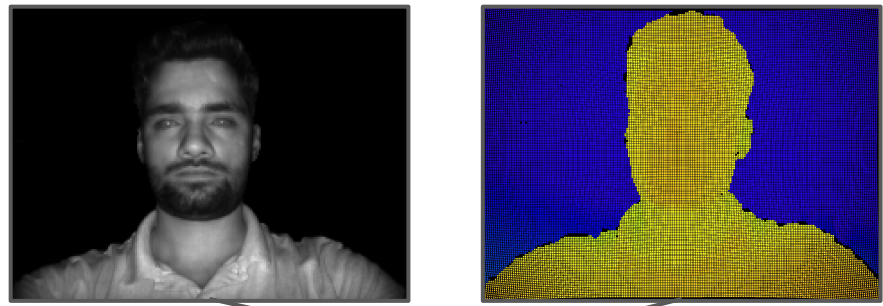
Results (Preliminary)

Understanding of ViT architecture.

To understand visualization of point cloud data using Open3d library.

- Captured a gray and depth image
- Merged (gray and depth images).

Depth map estimation.



Future Work

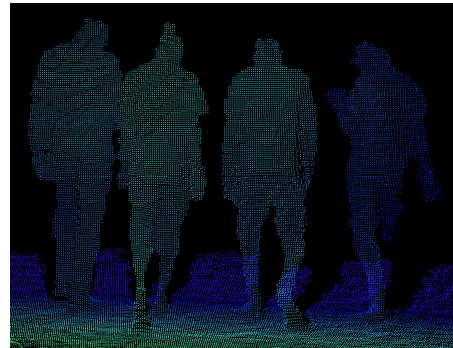
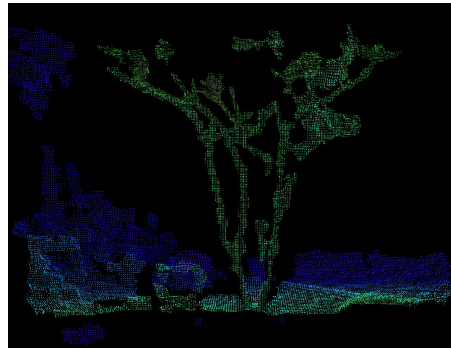
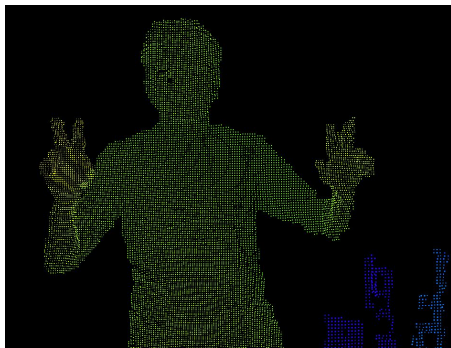
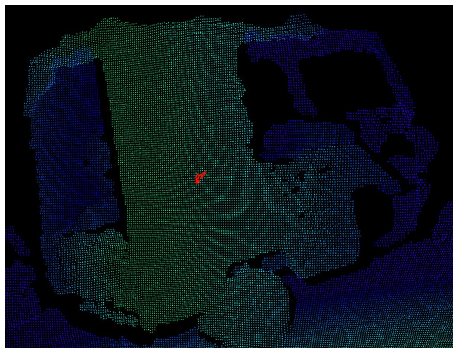
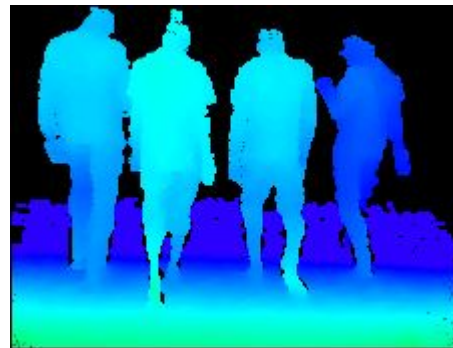
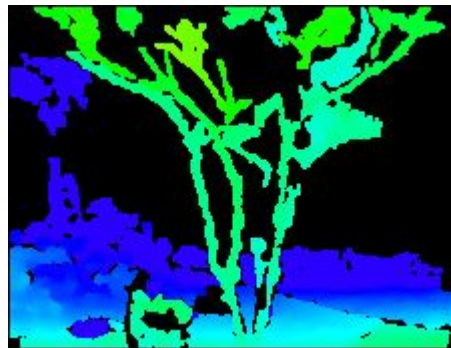
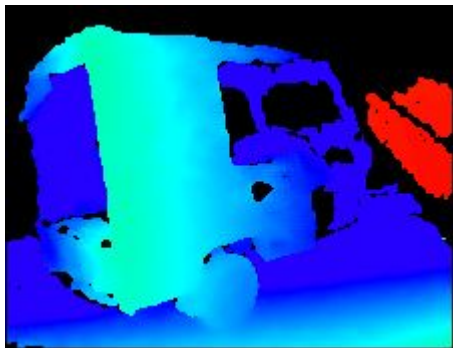
- Test and train existing Efficient Vit on Jetson Orin Nano.
- Implement a point transformer (PT) model.
- Understand the Torch Sparse library.
- Test Depth Anything model of existing dataset.
- Build an efficient vision point transformer.
- Compare depth estimation from PCD and DAM.

Thanks

Deep Learning (AI 5100)

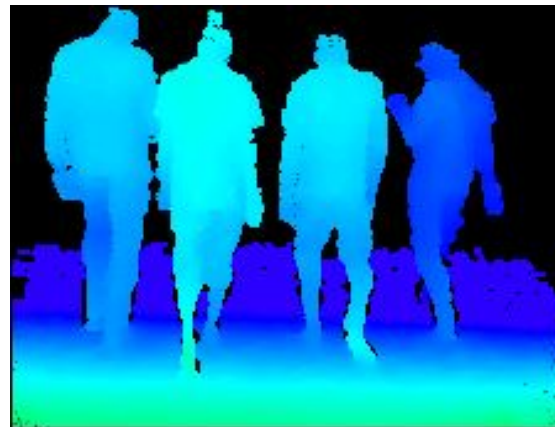
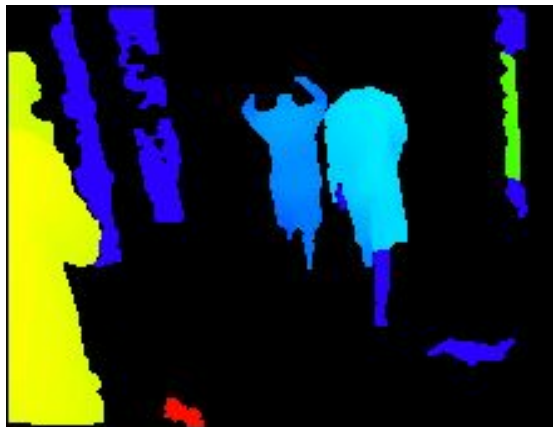
Project Progress

Depth Map and Point Cloud Data



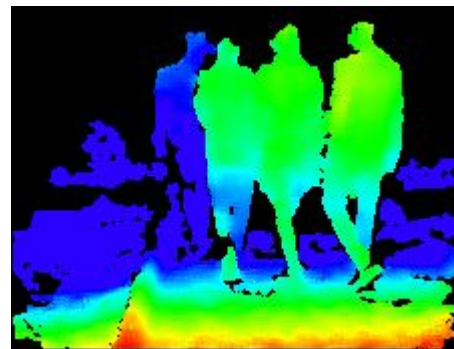
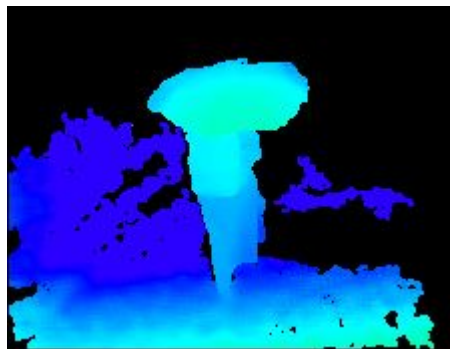
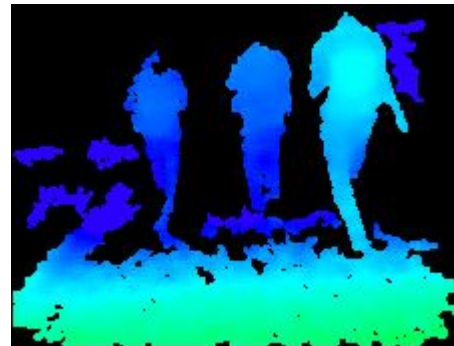
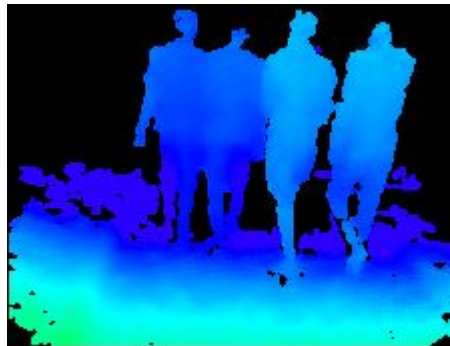
Sample video data

Data of people walking.



Sample data

Depth map of people walking



Segment Anything Model

Sample Gray scale Image



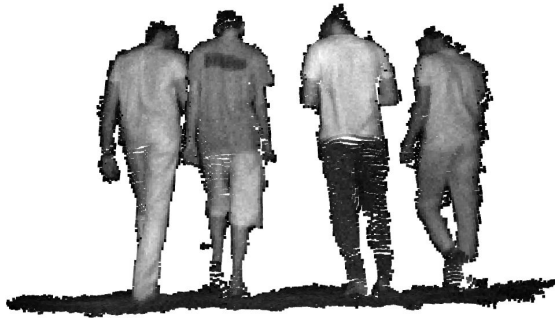
Segment Anything by Meta AI



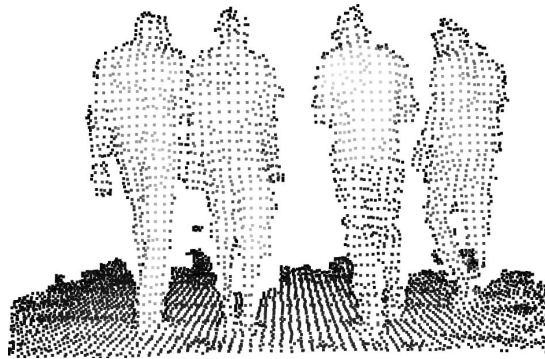
PCD Data Processing and Clustering



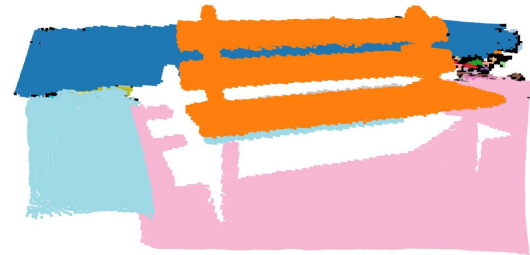
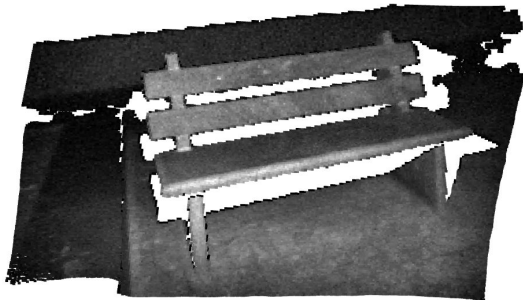
Visualize point cloud



Voxel Downsampling



DBSCAN Clustering



Efficient-Vit Benchmark on Jetson Nano



Benchmark

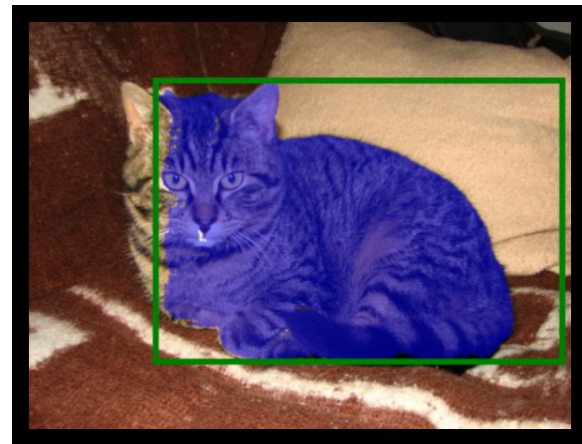
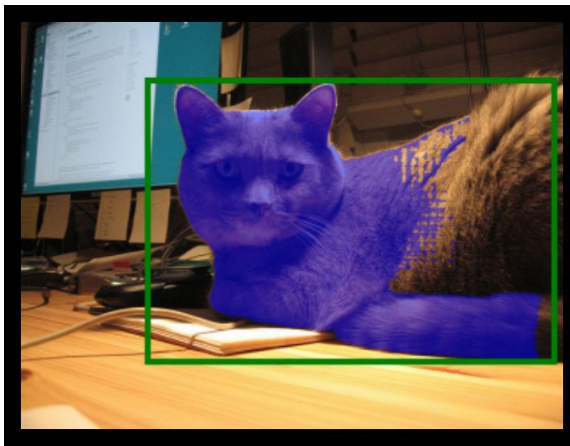
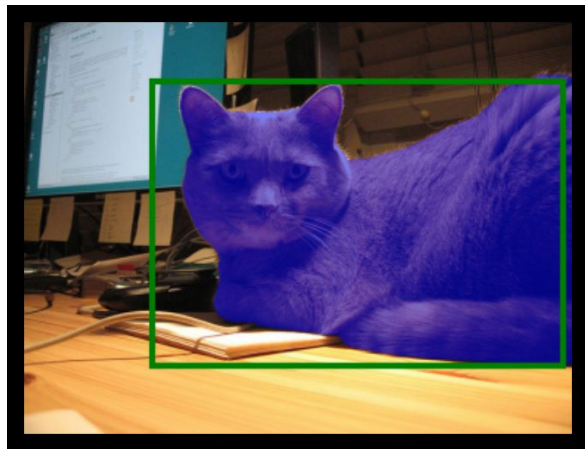
Test1

Test2

(Orin -64GB)

Orin - 8GB

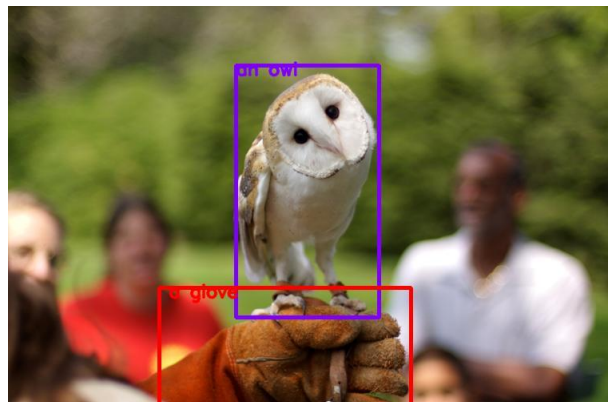
Orin - 8GB



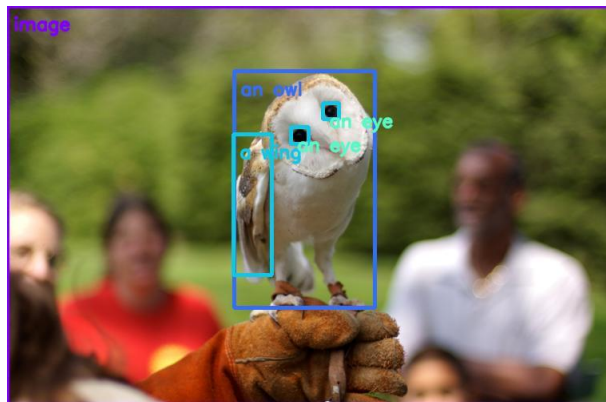
OWL-ViT (Open-World Localization Vision Transformer)

NanoOWL optimizes OWL-ViT to run in real-time 🔥 on NVIDIA Jetson Orin.

Basic Prediction



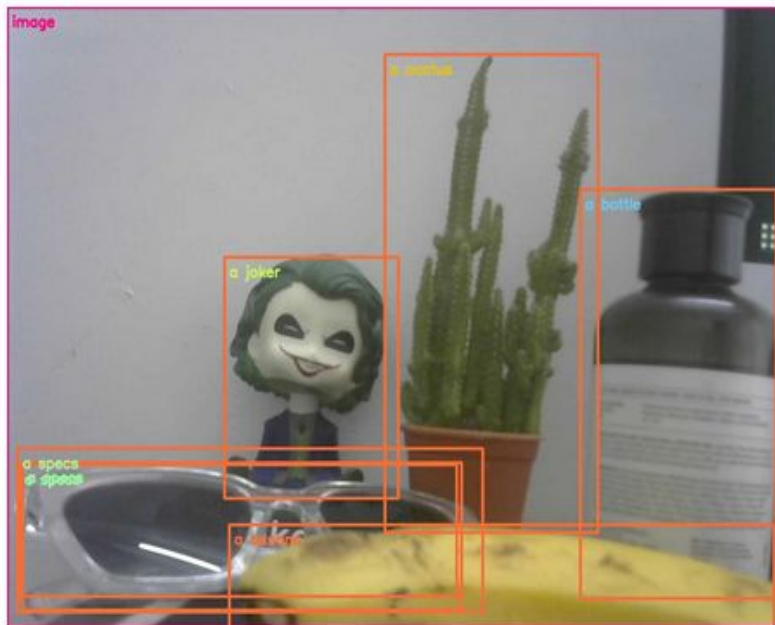
Tree Prediction



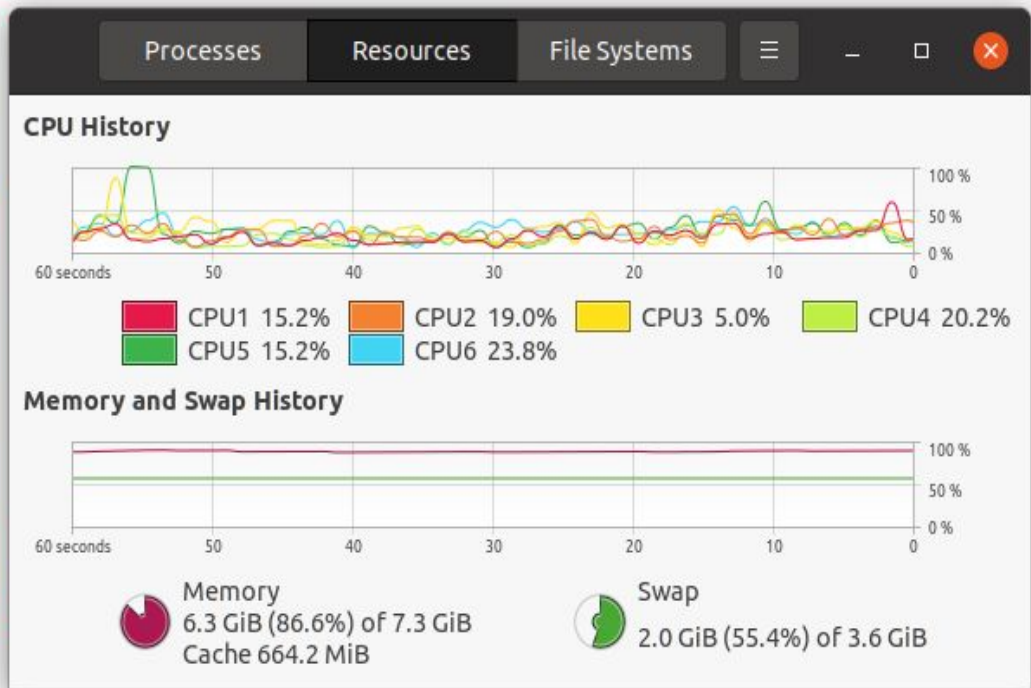
Real Time in Video



OWL-ViT (Demo) running on Jetson Orin Nano



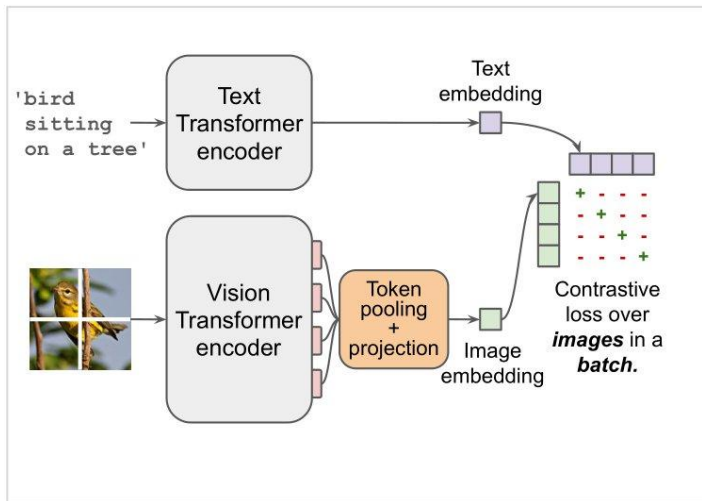
[a banana] [a cactus] [a joker] [a spoon] [a bottle]



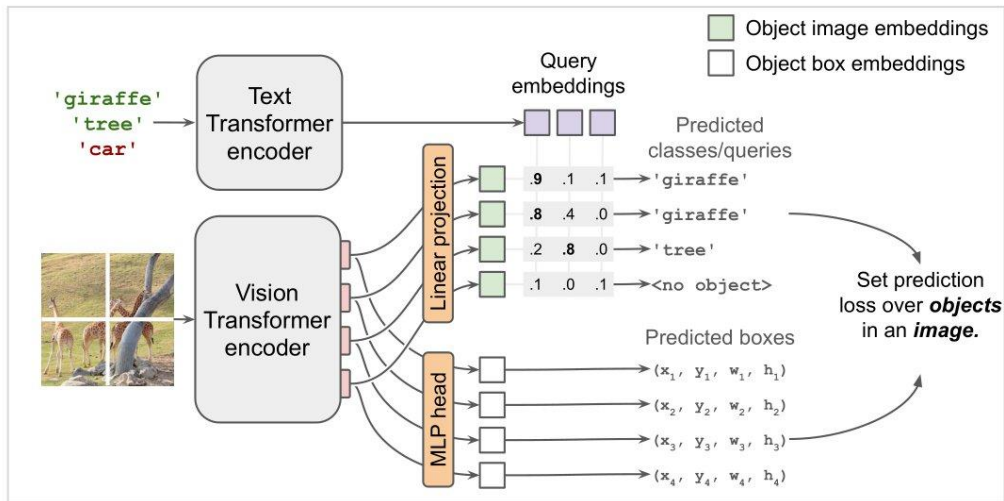
OWL-ViT

Architecture

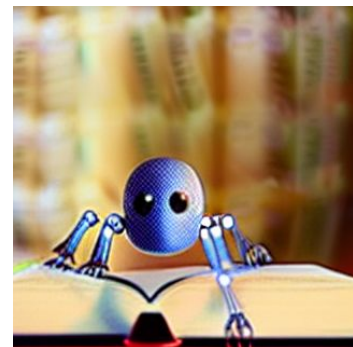
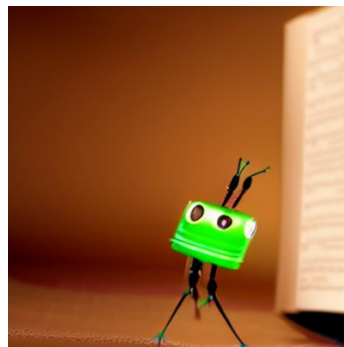
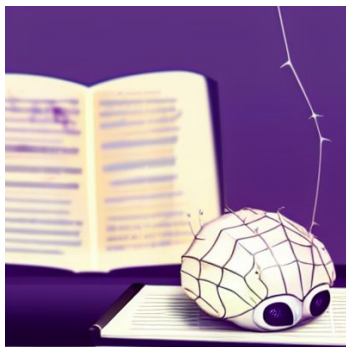
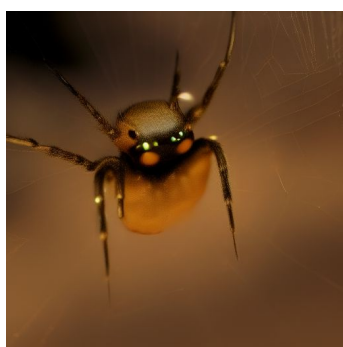
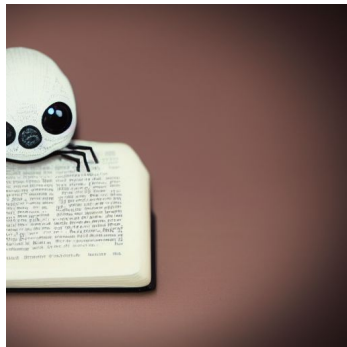
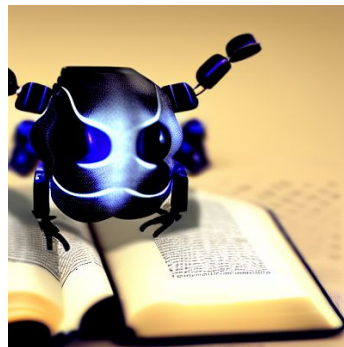
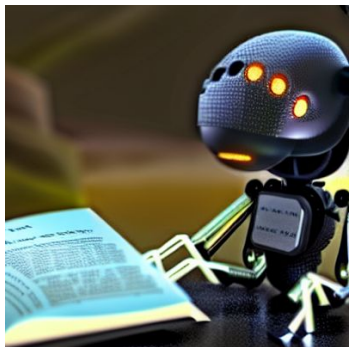
Image-level contrastive pre-training



Transfer to open-vocabulary detection



Stable Diffusion running on Jetson Nano



Prompt: A cute robotic spider with large artificial electrical brain reading a book.

Thanks